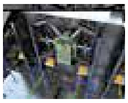
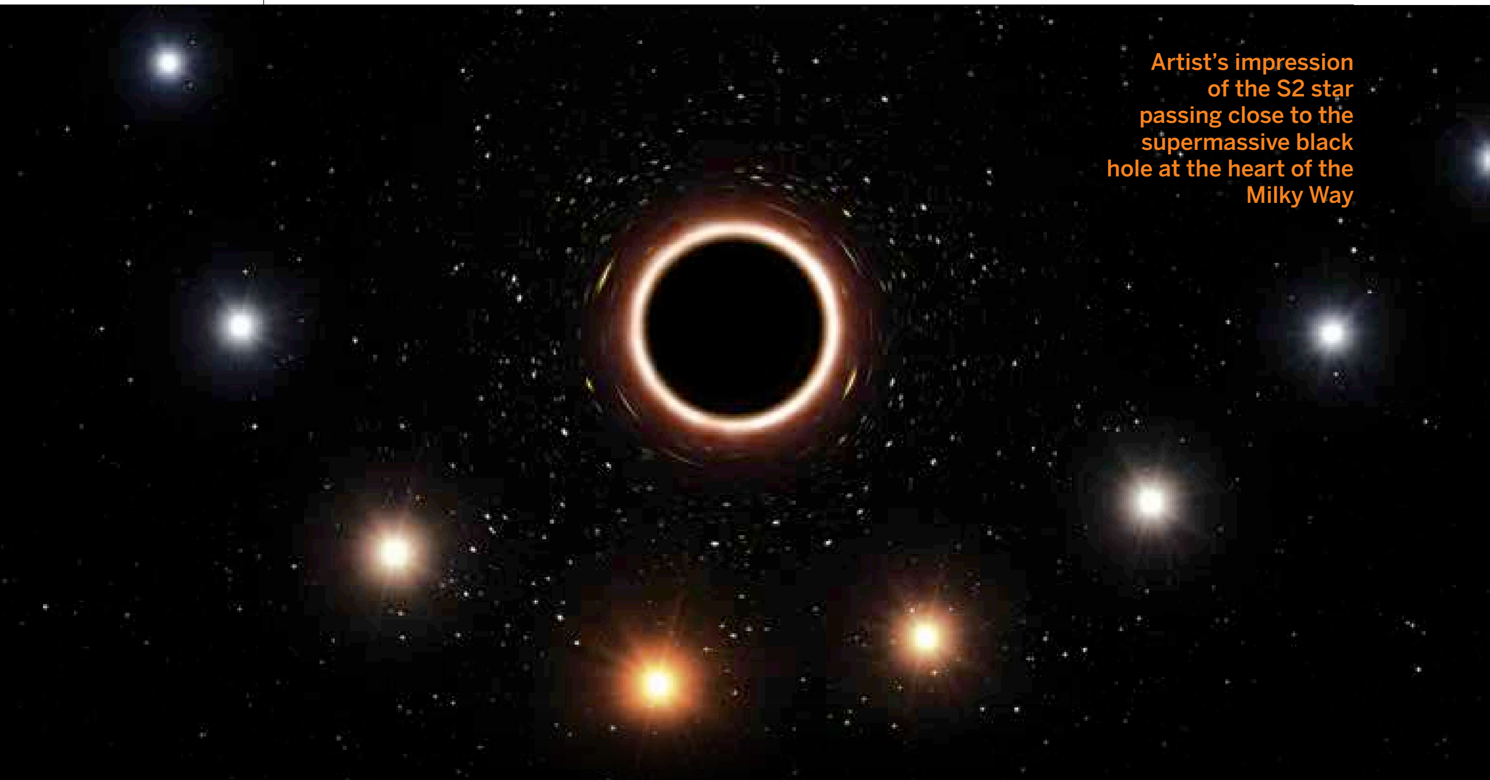


The path of light
Check out this animation of the incredibly complex path that light takes within the GRAVITY instrument. **https://dgit.in/apr20-01**

The other VLT instruments
There are a number of instruments at the ambitious VLT facility, here is a technical look at all of them. **https://dgit.in/apr20-02**



Artist's impression of the S2 star passing close to the supermassive black hole at the heart of the Milky Way

GRAVITY

The GRAVITY instrument is only getting started, allowing scientists to observe the cosmos in motion

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The Very Large Telescope (VLT) is made up of four primary or unit telescopes, and a number of auxiliary telescopes in the Atacama desert, Chile. It is operated by the European Southern Observatory. The telescopes can work individually, or in tandem with other instruments for the kind of observations necessary. The GRAVITY (not an acronym) instrument combines the near infrared light captured by all four telescopes, and combines them for observations. The light is micromanaged using polarisation filters, lasers and stretchable optical fibres to allow for the accuracy. It is extremely sensitive and can measure the exact distances to celestial objects with unprecedented precision. The precision of the instrument is comparable to pinpointing a five rupee coin on the surface of the Moon, within a centimeter of its actual position. This accuracy of the measurement allows scientists to track the movements of distant objects, on a daily basis. This allows observing the motion of everything from exoplanets to binary stars. It can be used to resolve small details in distant objects, or study objects obscured by the Milky Way. GRAVITY can be used to weigh or find out the masses of black holes, observe the exotic physics on their event horizons, study the accretion disks around young stars and investigate the hearts of active galactic nuclei.

SAGITTARIUS A*

Don't look for a footnote, the asterisk is part of the nomenclature. A monster lurks at the centre of the Milky Way, a blackhole that weighs as much as 4 million Suns. It is a strong radio source, one that was detected by Karl Jansky, a discovery that began the field of radio astronomy. You can read more about Jansky and his detection in the April 2019 issue of Digit, in Science, Ignorance. The radio source was designated Sagittarius A because of its proximity to the constellation. The centre of the galaxy is obscured by dust clouds in the visible spectrum, but scientists can peer through the obstacle in infrared. As things turned out, Sagittarius A was actually made up of thousands of stars very close together. In 1974, astronomers discovered a very massive and very

Credit: ESO/M Kornmesser